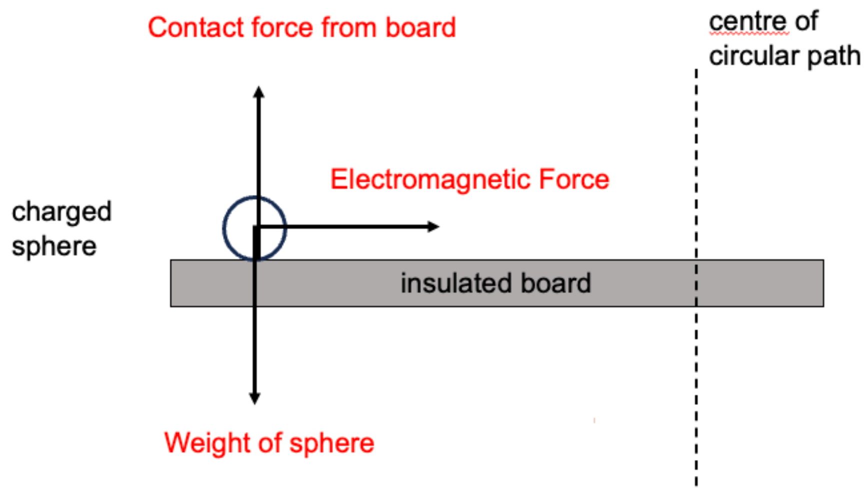
8	(a)	(i)	 1 mark per complete vector (with label)	**B3**
		(ii)	By Flemings left hand rule, vertically <u>downwards</u>	**B1**
		(iii)	Since centripetal acceleration is due to the electromagnetic force: $mr\omega^2 = Bqv = Bqr\omega$ $m\omega = Bq$ $m\frac{2\pi}{T} = Bq$ $T = \frac{2\pi m}{Bq}$ $T = \frac{2\pi(0.0060)}{0.5(5.0 \times 10^{-2})}$ $T = 1.5\ \mathrm{s}$	**M1** **M1** **A1**
		(iv)	With the electric field directed downwards, there is an <u>upward electric force</u>. <u>Hence the net vertical force is upwards.</u> Charged sphere experiences a upward acceleration concurrently with the centripetal acceleration. Charged particle starts to take <u>an upward helical path.</u>	**B1** **B1** **B1**

			Due to increasing upward velocity, the vertical distance between loops of the helical path increases.	
	(b)	(i)	Lenz's law of electromagnetic induction states that the direction of the induced e.m.f produced causes effect to oppose the change that produces it.	B1
		(ii)	As the rod slides across the rails, it will cut the magnetic flux. By Faraday's Law, there will be an e.m.f. induced. As the circuit is closed, there will be an induced current. OR As the rod rolls across the conducting rails, the area formed by the rod, rails and connecting wire is reduced. Hence there is a decrease in flux due to the decreasing area. By FL, there will be an induced e.m.f.. As the circuit is closed, there will be an induced current.	B1 B1
		(iii)	Using Faraday's Law, $E = Blv\sin\theta$ $E = (8.00)(0.500)(2.0)\sin 90$ $E = 8.0 \text{ V}$ $I = \frac{E}{R} = \frac{8.0}{10.0}$ $I = 0.80 \text{ A}$	M1 A1
		(iv)	By Fleming's left hand rule, the induced current in the rod and magnetic field results in a leftward EM force acting on the rod. This results in a net leftward force that decelerates the rod. The leftward EM force is dependent on the magnitude of induced current and therefore on the rod's velocity. Hence it decreases to zero as the rod slows down. Since the rod will come to a stop, there is no induced current thereafter and hence no leftward force. Therefore the rod will not gain leftward velocity. OR using energy concepts: The induced current in the rod and conducting rails requires energy for it to occur. This energy comes from the kinetic energy of the rod. Hence the rod loses KE and slows down. Once it comes to a complete stop, there is no more induced current.	B1 B1 B1 B1 B1 B1
		(v)	F_{ext} = Leftward EM force on rod due to induced current $F_{ext} = BIL\sin\theta$ $F_{ext} = (8.00)(0.80)(0.500)\sin 90$ $F_{ext} = 3.2 \text{ N}$	M1 A1